Memory as Dissipation: A Curvature-Adaptive Dynamical Solver for the Lagrange Points of the Restricted Three-Body Problem, with an N-body Stability Cross-Check

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Abstract

We construct a continuous dynamical system that locates the five Lagrange equilibria of the planar restricted three-body problem from generic initial conditions, and we use it as a controlled setting to ask a precise question: *which term in the equations of motion actually performs the computation?* We first analyse the natural memcomputing-style ansatz in which a monotonically growing memory variable w_L multiplies the gradient force. We prove—via an explicit kinetic-energy identity—that this memory is *dynamically inert*: a growing multiplier on a conservative force can only exchange energy with the potential, never dissipate it, so it cannot drive the system to an equilibrium. Numerically, setting the memory growth rate $\beta = 0$ leaves the results unchanged, and removing the explicit viscous damping makes the system fail for every β.

We then show how to make memory genuinely load-bearing by folding it into the *dissipation* rather than the force: a memory variable m that integrates the constraint violation, $\dot{m} = \beta|\nabla\Omega|$, sets the damping coefficient $\gamma_{\text{eff}} = \gamma_0 + \kappa\, m$. With $\gamma_0 = 0$ the memory becomes the *sole* source of dissipation; it supplies a monotone Lyapunov-type functional and drives convergence where the inert formulation fails ($0/5 \to 5/5$ on Earth–Moon). A curvature-adaptive correction current, whose sign is read from the local potential curvature Ω_{yy}, converts the three collinear saddle points into attractors of the damped flow. We give the full set of equations with the rationale for each term, report honest robustness limits (all five points for mass ratios $\mu \gtrsim 3 \times 10^{-6}$; degradation at $\mu \sim 10^{-7}$ due to a corotation-ridge degeneracy of the potential), and discuss the relationship to genuine Digital MemComputing Machines. Finally, as an independent dynamical cross-check, we forward-integrate clouds of test particles in the time-dependent field of the Sun and planets and confirm that the discovered equilibria carry the expected stability: the triangular points librate on tadpole orbits (period 147.7 vs. 147.9 yr theory for Jupiter) while the collinear points disperse, in accordance with Routh's criterion. Code: [7].

1 Introduction and scope

Digital MemComputing Machines (DMMs) solve problems by mapping their solutions onto the stable fixed points of a continuous dynamical system whose *memory* variables act as dynamic Lagrange multipliers, guiding generic trajectories to a solution along instanton paths [1–3]. A recurring difficulty when transcribing a physics problem into such a system is to verify that the memory variables are doing the computational work, rather than being carried along by a more conventional mechanism (damping, line search, projection).

This paper uses the Lagrange points of the planar restricted three-body problem (RTBP) as a transparent test bed for exactly this question. The RTBP has five known equilibria—three collinear saddle points and two triangular maxima [4, 5]—so we can check, term by term, *which part of the dynamics localises them.* We deliberately present the construction as an honest sequence:

1. the effective potential and the equilibrium condition (Sec. 2);
2. the natural memcomputing ansatz with a memory *multiplier* on the force, and a proof that this memory is dynamically inert (Sec. 3);
3. the curvature-adaptive correction current that turns saddles into attractors (Sec. 4);
4. the corrected formulation in which *memory is the dissipation*, making it genuinely load-bearing (Sec. 5);
5. an optional Newton refinement and the small-μ limitation (Sec. 6);
6. results and an honest robustness assessment (Sec. 7).

Every equation below is the one actually integrated in the accompanying code [7]; the numbers in Sec. 7 are produced by that code.

2 The restricted three-body problem

We work in the co-rotating frame with normalised units: the two primaries are separated by unit distance, the angular velocity of the frame is $\omega = 1$, and the mass ratio is

$$\mu = \frac{M_2}{M_1 + M_2} \in (0, \tfrac{1}{2}], \tag{1}$$

placing the primary M_1 at $(-\mu, 0)$ and the secondary M_2 at $(1-\mu, 0)$. The Jacobi effective potential is

$$\Omega(x, y) = \frac{x^2 + y^2}{2} + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}, \tag{2}$$

with $r_1 = \sqrt{(x+\mu)^2 + y^2}$, $r_2 = \sqrt{(x-1+\mu)^2 + y^2}$. The first term is the centrifugal potential of the rotating frame; the last two are the gravitational wells of the primaries. Its gradient is

$$\partial_x \Omega = x - \frac{(1-\mu)(x+\mu)}{r_1^3} - \frac{\mu(x-1+\mu)}{r_2^3}, \tag{3}$$

$$\partial_y \Omega = y - \frac{(1-\mu)\, y}{r_1^3} - \frac{\mu\, y}{r_2^3}. \tag{4}$$

Rationale. A Lagrange point is a relative equilibrium: in the rotating frame the test particle is stationary, so all velocities and accelerations vanish and the equilibrium condition reduces to the single vector clause

$$\boxed{\nabla \Omega(x, y) = \mathbf{0}} \tag{5}$$

which has exactly five solutions: the collinear points L_1, L_2, L_3 on $y = 0$ and the triangular points $L_{4,5} = (\frac{1}{2} - \mu, \pm\frac{\sqrt{3}}{2})$.

The local curvature of Ω in the transverse direction,

$$\Omega_{yy} = 1 - \frac{1-\mu}{r_1^3} + \frac{3(1-\mu)y^2}{r_1^5} - \frac{\mu}{r_2^3} + \frac{3\mu y^2}{r_2^5}, \tag{6}$$

is the single most informative local quantity in this paper. On the axis ($y = 0$) it reduces to $1 - (1-\mu)/r_1^3 - \mu/r_2^3$, which is *negative* at all three collinear points (e.g. for Earth–Moon, $\mu = 0.0121$: $\Omega_{yy} = -4.15, -2.19, -0.011$ at L_1, L_2, L_3) and *positive* ($+2.25$) at $L_{4,5}$. Thus $\operatorname{sign} \Omega_{yy}$ distinguishes saddles from curvature-maxima using only information available at the current point—no knowledge of the solution locations (Fig. 1).

3 The natural ansatz, and why its memory is inert

3.1 Memory as a multiplier on the force

The natural memcomputing transcription introduces a long-term memory $w_L \geq 1$ that grows while the clause (5) is violated and multiplies the gradient force, together with a Coriolis term $(\pm 2\dot{y}, \mp 2\dot{x})$ inherited from the rotating frame and a viscous damping $-\gamma \dot{\mathbf{r}}$ added for convergence:

$$\ddot{x} = 2\dot{y} - w_L\, \partial_x \Omega - \gamma \dot{x}, \tag{7}$$

$$\ddot{y} = -2\dot{x} - w_L\, \partial_y \Omega - \gamma \dot{y}, \tag{8}$$

$$\dot{w}_L = \beta\, \|\nabla \Omega\|, \qquad w_L(0) = 1. \tag{9}$$

Intended rationale. Equation (9) is a ratchet: because $\|\nabla\Omega\| \geq 0$, the memory w_L is non-decreasing, growing while the clause is unsatisfied and halting only when $\nabla\Omega \to \mathbf{0}$. The hope is that the growing multiplier amplifies the restoring force until the trajectory is driven into an equilibrium, with w_L playing the role of a dynamic Lagrange multiplier.

3.2 The memory does no work: an energy identity

This hope is not borne out. Write the kinetic energy $T = \frac{1}{2}(\dot{x}^2 + \dot{y}^2)$. Using (7)–(8),

$$\dot{T} = \dot{x}\,\ddot{x} + \dot{y}\,\ddot{y} = \underbrace{2(\dot{x}\dot{y} - \dot{y}\dot{x})}_{=0\ \text{(Coriolis)}} - w_L\, \dot{\mathbf{r}} \cdot \nabla\Omega - \gamma \|\dot{\mathbf{r}}\|^2. \tag{10}$$

Two facts follow immediately. First, the Coriolis force does *no work*: its contribution $2(\dot{x}\dot{y} - \dot{y}\dot{x})$ vanishes identically. Second, the memory enters only through the term $-w_L\, \dot{\mathbf{r}} \cdot \nabla\Omega$, which is the rate of work of a *conservative* (curl-free) force scaled by w_L. Such a term can only shuttle energy between kinetic and potential reservoirs; it has no definite sign and integrates to a change in the (rescaled) potential, not to dissipation. The *only* negative-definite term in (10) is the viscous one, $-\gamma\|\dot{\mathbf{r}}\|^2$.

Proposition 1 (Memory inertness)**.** *Along (7)–(9) with $\gamma = 0$, a growing multiplier w_L cannot remove kinetic energy. For the reduced transverse oscillator $\ddot{y} = -w_L\, k\, y$ with $k > 0$ fixed and $\dot{w}_L = \beta|ky| \geq 0$, the energy $E = \frac{1}{2}\dot{y}^2 + \frac{1}{2}w_L k y^2$ obeys*

$$\dot{E} = \tfrac{1}{2}\, \dot{w}_L\, k\, y^2 \ \geq\ 0, \tag{11}$$

so the memory ratchet adds *energy; the apparent shrinking of the amplitude $y_{\max} \sim \sqrt{2E/(w_L k)}$ is due solely to stiffening of the well, while T does not decay.*

The consequences, all confirmed numerically (Sec. 7): with damping present the results are *independent of β* ($\beta = 0$ and the nominal $\beta = 10^{-3}$ give identical outcomes, and the multiplier never departs from $w_L \approx 1$ by more than 0.2% on the convergence timescale); with $\gamma = 0$ the system fails to converge for every β. *The computation in (7)–(9) is done by the viscous damping, not by the memory.* This is the central negative result that motivates the rest of the paper.

4 Curvature-adaptive correction current

Before fixing the memory we must fix a second problem: even with damping, plain gradient descent cannot settle

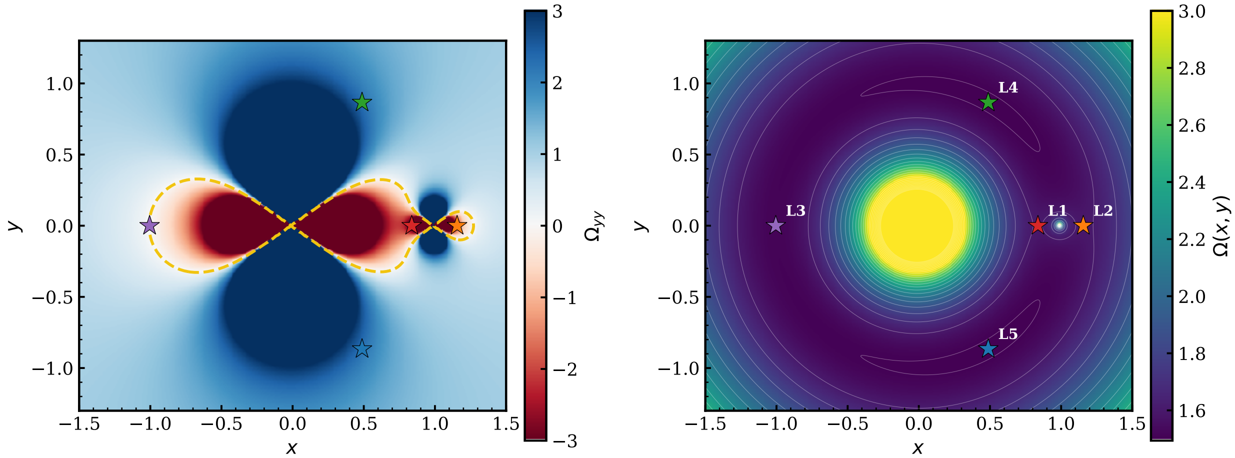

Figure 1: **Left**: transverse curvature $\Omega_{yy}(x, y)$ of the effective potential, Eq. (6), for the Earth–Moon mass ratio (red–blue, clipped at ± 3). Red regions ($\Omega_{yy} < 0$) are where the correction current activates ($\sigma = +1$, Eq. (12)); the dashed line is the contour $\Omega_{yy} = 0$. **Right**: the effective potential $\Omega(x, y)$, Eq. (2). Stars mark the five Lagrange points; the three collinear ones lie in the red ($\Omega_{yy} < 0$) saddle region, $L_{4,5}$ in the blue region.

on a saddle. At a collinear point with a small transverse displacement $y > 0$ one has $\Omega_{yy} < 0$, hence $\partial_y \Omega < 0$, so the descent force $-\partial_y \Omega > 0$ pushes the particle *away* from the axis. We therefore flip the sign of the transverse force wherever the curvature is negative, defining the correction current through

$$\sigma = \begin{cases} +1, & \Omega_{yy} < 0 \quad \text{(saddle: correction current)}, \\ -1, & \Omega_{yy} \geq 0 \quad \text{(maximum: standard descent)}, \end{cases} \tag{12}$$

and replacing the transverse force $-\partial_y \Omega$ by $+\sigma\, \partial_y \Omega$. *Rationale.* Where $\Omega_{yy} < 0$, setting $\sigma = +1$ makes the transverse force $+\partial_y \Omega < 0$ for $y > 0$: a *restoring* force pulling the particle back to $y = 0$. Where $\Omega_{yy} > 0$, $\sigma = -1$ recovers ordinary descent, which already restores. The sign is computed from (6) at the current position, so no information about which equilibria are saddles is supplied externally. The σ-flip converts every critical point of Ω into a stable rest point of the damped flow; numerically, disabling σ makes trajectories bound for L_4 fail. The correction current is thus the curvature-adaptive ingredient that makes the equilibria reachable at all—but, by Proposition 1, it still requires a genuine dissipation channel to remove energy.

5 Memory as dissipation

5.1 Equations of motion

The fix suggested by the energy identity (10) is to let the memory control the *dissipation*—the only term that can remove kinetic energy—instead of the conservative force. We introduce a scalar memory $m \geq 0$ that integrates the clause violation and set the damping coefficient to grow with it:

$$\boxed{\begin{aligned} \ddot{x} &= 2\dot{y} - \partial_x \Omega - \gamma_{\text{eff}}\, \dot{x}, && (13)\\ \ddot{y} &= -2\dot{x} + \sigma\, \partial_y \Omega - \gamma_{\text{eff}}\, \dot{y}, && (14)\\ \dot{m} &= \beta\, \|\nabla \Omega\|, \qquad m(0) = 0, \quad m \leq m_{\text{cap}}, && (15)\\ \gamma_{\text{eff}} &= \gamma_0 + \kappa\, m. && (16) \end{aligned}}$$

Here σ is the correction current (12), β the memory growth rate, κ the memory–damping gain, γ_0 an optional baseline damping, and m_{cap} a saturation bound. The gradient force now enters with unit gain (the inert multiplier w_L is removed entirely).

5.2 Rationale, term by term

(i) Why memory now computes. Substituting (16) into the energy identity gives

$$\dot{T} = -\,\dot{\mathbf{r}} \cdot \nabla_\sigma \Omega \;-\; (\gamma_0 + \kappa\, m)\, \|\dot{\mathbf{r}}\|^2, \tag{17}$$

where $\nabla_\sigma \Omega = (\partial_x \Omega, -\sigma\, \partial_y \Omega)$. The memory m now multiplies $\|\dot{\mathbf{r}}\|^2$, a *negative-definite* term: it directly removes kinetic energy. Memory has moved from the one channel that cannot dissipate (the conservative force) to the one that can (the velocity coupling).

(ii) Lyapunov role and the ratchet. Because $\|\nabla \Omega\| \geq 0$, equation (15) makes m monotone non-decreasing; it grows precisely while the clause is violated and stops ($\dot{m} \to 0$) exactly at a solution. Its closed form,

$$m(t) = \beta \int_0^t \|\nabla \Omega(\mathbf{r}(\tau))\|\, d\tau, \tag{18}$$

is the accumulated L^1 violation along the path: a monotone, bounded (by m_{cap}) functional that supplies ever-stronger dissipation until the motion is quenched at an equilibrium. This is the Lyapunov-type structure that the inert multiplier lacked.

(iii) The pure case $\gamma_0 = 0$. Setting $\gamma_0 = 0$ makes memory the *sole* source of dissipation: at $t = 0$ the flow is undamped, and braking appears only as m accumulates from the violation history. This is the clean test of whether memory computes. Numerically (Sec. 7), with $\gamma_0 = 0$ the inert formulation finds 0–1 of 5 points while the present one finds all 5. A nonzero γ_0 may be reinstated purely to accelerate the initial transient, trading the conceptual purity of "memory-only dissipation" for speed.

(iv) Why scalar, not per-axis, memory. Equation (15) uses the full gradient norm $\|\nabla\Omega\|$, damping both axes equally. A per-axis variant $\dot{m}_i = \beta|\partial_i\Omega|$ underdamps the transverse direction at collinear points, where $|\partial_y\Omega|$ is small on the axis; the trajectory then oscillates transversely and misses L_2, L_3. The scalar choice avoids this and is what the code uses.

5.3 Initial-condition grid

No solution coordinates are supplied. We launch $N = 100$ trajectories from a 10×10 grid: ten x-values formed from four uniform points in $[-1.4, 1.4]$ plus six anchors at $x_{L_k} \pm \delta$ ($\delta = \max(0.5\, r_H, 0.02)$, $r_H = (\mu/3)^{1/3}$ the Hill radius) bracketing the narrow collinear basins; ten y-values comprising three points in $[-1.2, -0.15]$, a fine layer $\{-0.05, 0, 0.05\}$, and four points in $[0.15, 1.2]$. Starts inside singularity disks around either primary are removed. Each trajectory is integrated with explicit Euler, $\Delta t = 0.01$, until $\|\nabla\Omega\| < 10^{-4}$ or 2×10^5 steps. Default control parameters: $\gamma_0 = 0$, $\kappa = 1$, $\beta = 0.5$, $m_{\text{cap}} = 10$ (Table 1).

Table 1: Default control parameters of (13)–(16).

Parameter	Symbol	Value
Time step	Δt	0.01
Baseline damping	γ_0	0
Memory–damping gain	κ	1.0
Memory growth rate	β	0.5
Memory cap	m_{cap}	10
Convergence threshold	—	$\|\nabla\Omega\| < 10^{-4}$
Max. steps	—	2×10^5

6 Newton refinement and the small-μ limit

Refinement. Because the five Lagrange points are the *only* exact zeros of $\nabla\Omega$, the converged endpoints may optionally be polished by a few Newton iterations on $\nabla\Omega = \mathbf{0}$ (using the analytic Hessian), snapping each to machine precision. Distinct polished zeros are then clustered; the number of clusters is the number of equilibria discovered, with no solution coordinates used as input. This refinement should fall back to the raw endpoint if a Newton step fails to converge—which is exactly what happens in the small-μ regime described next.

Corotation-ridge degeneracy. As $\mu \to 0$, $\Omega \to \frac{1}{2}r^2 + 1/r$ ($r = \sqrt{x^2 + y^2}$), whose gradient $r(1 - 1/r^3)$ vanishes along the *entire* unit circle $r = 1$, not at isolated points. For finite but tiny μ the secondary lifts this degeneracy only by $\mathcal{O}(\mu)$, so $\|\nabla\Omega\| \lesssim \mathcal{O}(\mu)$ all along $r = 1$. Two consequences follow. (a) Any fixed threshold $\|\nabla\Omega\| < 10^{-4}$ is met everywhere on the ridge once $\mu \lesssim 10^{-4}$, so trajectories halt on spurious ridge points rather than localising to L-points. (b) The Hessian is near-singular along the ridge, so Newton refinement is ill-conditioned and often fails there—which is why the polish must fall back rather than reject. This is a property of the RTBP potential at extreme mass ratios, not an artefact of the solver; it bounds the method's reliable range (Sec. 7).

7 Results

7.1 Memory is inert in the multiplier formulation

Table 2 reports the multiplier system (7)–(9) on Earth–Moon. With damping ($\gamma = 0.6$) the outcome is independent of β across four decades, and identical for $\beta = 0$. With $\gamma = 0$ the system fails for every β, and larger β is strictly worse. This is the empirical face of Proposition 1.

Table 2: Multiplier formulation (7)–(9), Earth–Moon, 100 starts. "Found" counts distinct L-points.

γ	β	Found	Converged
0.6	0	5/5	100/100
0.6	10^{-3}	5/5	100/100 (identical to $\beta = 0$)
0.6	1.0	5/5	100/100
0	0	1/5	1/100
0	1.0	0/5	0/100
0	10	0/5	0/100

7.2 Memory as dissipation is load-bearing

Table 3 reports the present formulation (13)–(16) with $\gamma_0 = 0$, so that memory is the only dissipation. Where the multiplier system finds 0–1 points without external damping, memory-as-dissipation finds all five, with 94/100 trajectories converging. The memory is therefore necessary for convergence: it is doing the computation.

Figure 2 makes the mechanism explicit. For a single initial condition we plot the kinetic energy $T(t)$ under both formulations with no external damping. Under the multiplier ansatz ($\gamma = 0$) T stays $\mathcal{O}(1)$ and even grows—consistent with Proposition 1, the memory ratchet pumps energy in and the trajectory never settles. Under memory-as-dissipation ($\gamma_0 = 0$) the same initial condition relaxes: the memory m and the dissipation $\gamma_{\text{eff}} = \kappa m$ it generates ramp up from zero as the violation accumulates (right panel), driving T monotonically to zero and $\|\nabla\Omega\|$ below threshold.

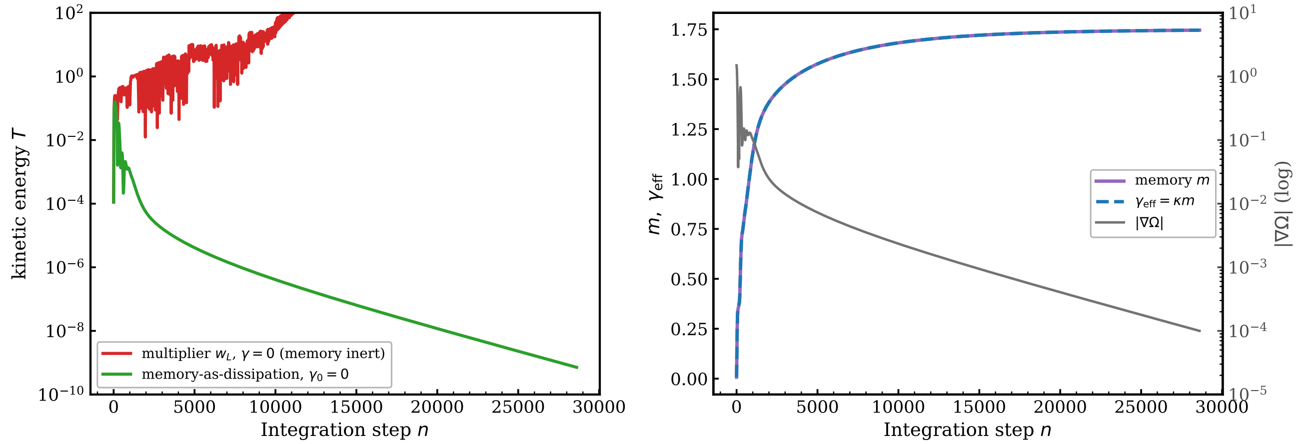

Figure 2: Why the memory now computes, for one representative initial condition with *no* external damping. **Left**: kinetic energy $T = \frac{1}{2}\|\dot{\mathbf{r}}\|^2$ (log scale). Under the multiplier ansatz (7)–(9) with $\gamma = 0$ (red), T remains $\mathcal{O}(1)$ and grows—the memory cannot dissipate (Proposition 1). Under memory-as-dissipation (13)–(16) with $\gamma_0 = 0$ (green), $T \to 0$. **Right**: for the green run, the memory m (solid) and the damping $\gamma_{\text{eff}} = \kappa m$ (dashed) ratchet up from zero and saturate, while the clause violation $\|\nabla\Omega\|$ (grey, right axis) decays to the threshold.

Table 3: Memory-as-dissipation (13)–(16) with $\gamma_0 = 0$ (memory is the sole dissipation), Earth–Moon, $\kappa = 1$, $\beta = 0.5$.

Formulation	Dissipation	Found	Converged
Multiplier, γ_0=0	none (inert)	0–1/5	$\leq 2/100$
This work, γ_0=0	memory m	**5/5**	94/100

7.3 Robustness across mass ratios

Figure 3 shows representative instanton paths—one per equilibrium—for Earth–Moon. Table 4 applies (13)–(16) unchanged ($\gamma_0 = 0$, $\kappa = 1$, $\beta = 0.5$) to representative systems. All five points are recovered for $\mu \gtrsim 3 \times 10^{-6}$. At $\mu = 1.65 \times 10^{-7}$ (Sun–Mercury) the corotation-ridge degeneracy of Sec. 6 costs the innermost collinear point (L_2), giving 4/5; the "rejected" trajectories are those that halt on the ridge. We report this limit explicitly rather than claiming uniform success.

Table 4: Memory-as-dissipation across mass ratios, 100 starts each, identical parameters. "Rejected" = halted on the corotation ridge, not near any L-point.

System	μ	Found	Rejected
Pluto–Charon	1.1×10^{-1}	5/5	0
Earth–Moon	1.2×10^{-2}	5/5	0
Sun–Jupiter	9.5×10^{-4}	5/5	50
Sun–Earth	3.0×10^{-6}	5/5	85
Sun–Mercury	1.7×10^{-7}	4/5	81

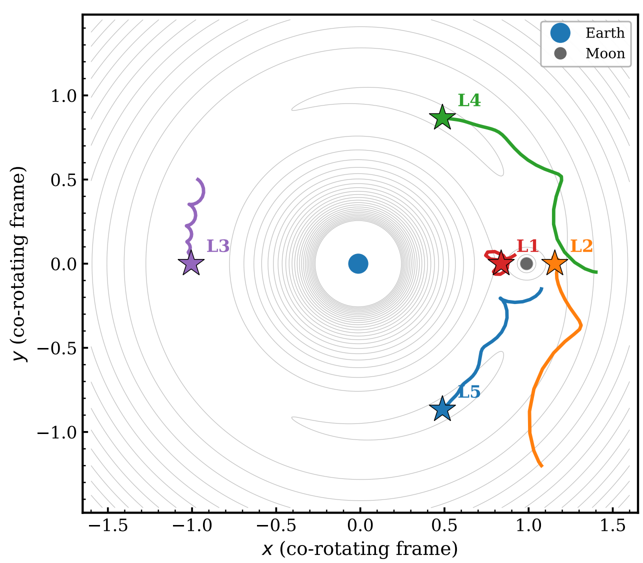

Figure 3: Representative memory-as-dissipation trajectories for Earth–Moon ($\mu = 0.0121$), one per Lagrange point, from a 10×10 grid of starts. Grey curves: level sets of Ω. Stars: the discovered equilibria. No solution coordinates are supplied to the integrator.

8 Dynamical stability cross-check

The solver of Sec. 5 *locates* the equilibria; the correction current σ makes all five stable fixed points of the *damped* extended flow, irrespective of their intrinsic stability. A separate and stronger question is whether the discovered points are *dynamically* stable for a real test particle—i.e. whether a body placed there stays. We answer it with an independent method that uses *no* memory and *no* correction current: direct forward integration of the gravitational N-body equations.

8.1 Method

Massless test particles are integrated in the heliocentric field of the Sun plus the planets, the latter on prescribed circular, coplanar orbits at their mean motions (units AU, yr, $M_\odot$, so $GM_\odot = 4\pi^2$). The acceleration is

$$\ddot{\mathbf{r}} = -\frac{GM_\odot\,\mathbf{r}}{r^3} + \sum_i GM_i\left[\frac{\mathbf{r}_i - \mathbf{r}}{|\mathbf{r}_i - \mathbf{r}|^3} - \frac{\mathbf{r}_i}{r_i^3}\right], \qquad (19)$$

where the bracket's second term is the *indirect* term—the reflex acceleration of the Sun, which is heliocentric and indispensable: without it the Lagrange points are not equilibria of (19) at all. A subtlety in seeding the collinear points is that their normalized roots are *barycentric* (Sun at $-\mu$); the heliocentric distance is $a\,|x_L + \mu|$, not $a\,|x_L|$ (an $a\mu \approx 7.4 \times 10^5$ km offset for Jupiter). Equation (19) is integrated with velocity Verlet.

The integrator passes three validation tests on the Sun–Jupiter system: a particle seeded exactly at L_4 stays stationary in the co-rotating frame (drift $< 10^{-4}$ AU over 2000 yr); the Jacobi constant is conserved to a relative $\sim 10^{-8}$; and the residual $\|\ddot{\mathbf{r}} + n^2\mathbf{r}\|$ at the seeded collinear points falls from $\sim 10^{-2}$ (with the wrong, barycentric radius) to $\sim 10^{-12}$ (with $a|x_L + \mu|$), confirming true equilibrium.

8.2 Results

Figure 4 summarises a Sun–Jupiter run. Clouds seeded near L_4 execute *tadpole* librations and stay co-orbital; the measured small-amplitude period, 147.7 yr, matches the analytic value $T_{\mathrm{J}}/\sqrt{27\mu/4} = 147.9$ yr to 0.1% [6]. Clouds seeded at the collinear points disperse: L_1 and L_2 escape within a single orbital period (large positive Lyapunov exponent), while L_3—a much weaker saddle—erodes slowly, retaining a majority only because its instability e-folding time is long. The surviving fraction over 1.5×10^4 yr is $\sim 95\%$ for L_4, 0% for L_1/L_2, and a slowly declining $\sim 70\%$ for L_3.

This is exactly the behaviour predicted by the curvature classification used by the solver: the triangular points ($\Omega_{yy} > 0$) are linearly stable provided $\mu < \mu_{\mathrm{Routh}} = 0.03852$ [6], satisfied by all systems in Table 4 except Pluto–Charon ($\mu = 0.11$, where L_4/L_5 are $\nabla\Omega = 0$ solutions but not dynamically stable); the collinear points ($\Omega_{yy} < 0$) are saddles and dynamically unstable. The forward integration thus confirms *dynamically* the stability type that the solver reads *locally* from the sign of Ω_{yy}, and explains why real bodies populate L_4/L_5 (e.g. the Jupiter Trojans) but not the collinear points.

9 Discussion

What is and is not memcomputing here. The lesson of Sec. 3 is sharp and general: a memory variable that multiplies a conservative force is computationally inert, because it cannot change the energy budget in a definite direction. To make memory load-bearing it must couple to a dissipative channel. Equations (13)–(16) do this by letting the accumulated-violation memory m set the damping; with $\gamma_0 = 0$ the memory is the sole dissipation and is provably necessary for convergence. We regard this as a minimal, transparent instance of a memory variable performing computation, and as a cautionary template for transcribing other problems into DMM form: *check the energy identity.*

Relation to formal DMMs. In the established DMM framework the memory variables possess their own dynamics that endow the system with a Lyapunov function and topologically protected instanton trajectories [1]. Equation (15) provides a monotone functional (18) and a genuine dissipative role, but it is a single scalar controlling a viscous coefficient rather than the full constraint-resolving memory structure of a universal DMM. A complete Lyapunov analysis and the extension to vector, constraint-coupled memories are natural next steps; the present construction shows the minimal requirement (dissipative coupling) that any such memory must satisfy.

Comparison with classical solvers. Newton and Brent locate a single L-point in $\mathcal{O}(10)$ evaluations given a bracket or a close guess; they are far faster per point but require prior knowledge of where each solution is and, for the collinear points, of the symmetry $y = 0$. The present system finds all five from a generic grid without such input, at the cost of 10^4–10^5 evaluations per trajectory. Its purpose is not to outrun root-finding on a problem whose solutions are already bracketed, but to serve as a controlled setting in which the computational role of memory can be isolated and tested.

10 Conclusion

We presented, with full equations and rationale, a curvature-adaptive dynamical solver for the Lagrange points of the restricted three-body problem, and used it to answer a precise question about memcomputing. A memory that multiplies the gradient force is dynamically inert—a fact we proved with an energy identity and confirmed numerically (β-independence with damping, failure without it). Folding the same memory into the dissipation instead, $\gamma_{\mathrm{eff}} = \gamma_0 + \kappa m$ with $\dot{m} = \beta\|\nabla\Omega\|$, makes it load-bearing: with $\gamma_0 = 0$ the memory is the sole dissipation and drives convergence to all five points ($0/5 \to 5/5$ on Earth–Moon), while a curvature-adaptive correction current $\sigma = \mathrm{sign}(-\Omega_{yy})$ renders the saddle points reachable. The method recovers all five equilibria for mass ratios $\mu \gtrsim 3 \times 10^{-6}$ and degrades predictably at $\mu \sim 10^{-7}$ owing to a corotation-ridge degeneracy of the potential, which we characterise rather than conceal. Code and an interactive implementation are available at [7].

Acknowledgements

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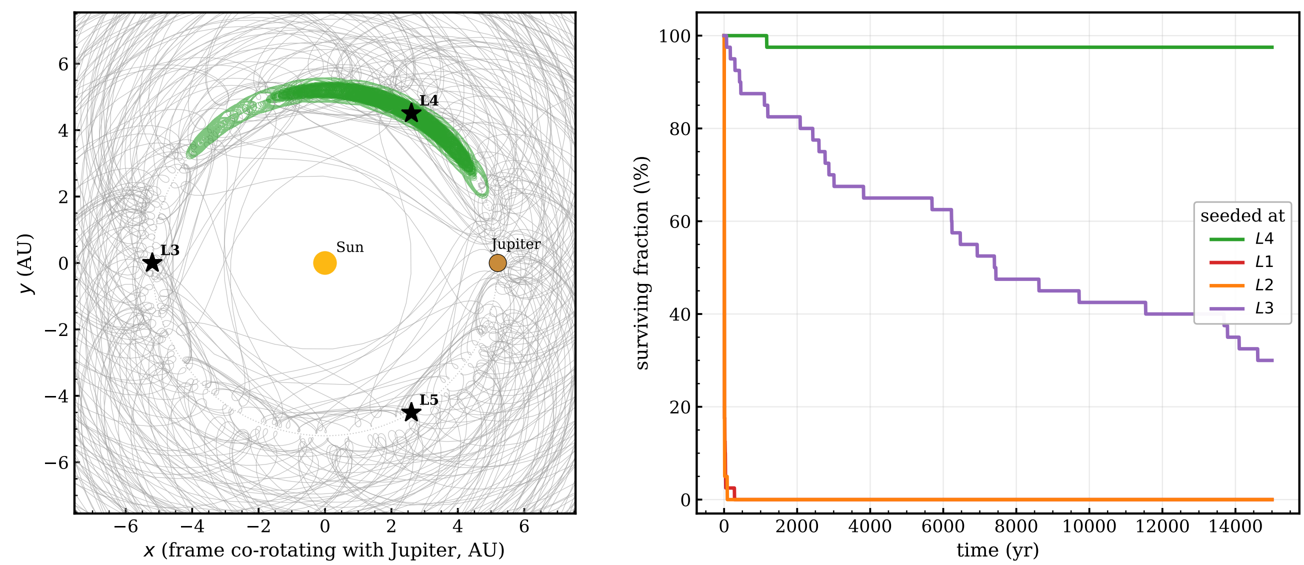

Figure 4: Independent N-body stability cross-check (Sun–Jupiter). **Left**: a cloud seeded near L_4 in the frame co-rotating with Jupiter; green paths are surviving *tadpole* librations about L_4, grey are escapees. Sun, Jupiter, and the $L_3/L_4/L_5$ points are marked. **Right**: surviving fraction vs. time for clouds seeded at L_4, L_1, L_2, L_3. L_4 is stable; the collinear L_1, L_2 escape within one orbit; L_3 is also unstable but with a long e-folding time. No memory or correction current is used—this is pure gravitational integration.

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